

A Strategic Decision Support System for Customer Churn Prediction

Phase 2 Final Report

Dokuz Eylul University | Graduate Program in Computer Engineering

Course: Strategic Information Systems

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Abstract

Customer churn is a strategic information systems problem because it affects revenue stability, retention spending, customer lifetime value, and analytical CRM capability. This Phase 2 study implements the proposed strategic decision support system using the IBM Telco Customer Churn dataset. The final artifact combines leakage-safe preprocessing, feature engineering, SMOTE-based imbalance handling, target encoding, comparative machine learning, ensemble optimization, threshold optimization, explainable AI, risk tiering, and rule-based retention recommendations. The selected operational model is Focused Voting Ensemble, evaluated on a stratified hold-out test set with accuracy 0.813, ROC-AUC 0.849, and PR-AUC 0.668 at the standard 0.50 threshold. The accuracy-oriented threshold of 0.51 raises hold-out accuracy to 0.815, while the cost-sensitive threshold of 0.14 raises recall to 0.912 for retention-oriented decision support.

1. Introduction

In saturated telecommunications markets, service bundles are increasingly imitable and switching costs are often low. Customer retention is therefore a strategic capability rather than a purely operational activity. Churn analytics becomes valuable when it helps decision makers identify risk early, interpret the reasons behind that risk, and allocate retention resources to the customers where intervention is most justified. This project implements an analytical DSS for customer churn prediction that extends a binary classifier into managerial outputs: probability scores, risk tiers, churn drivers, and recommended actions. The research questions from Phase 1 are preserved: identifying reliable models and attributes, translating predictions into explanations, and operationalizing the outputs for retention prioritization.

2. Literature Review

Recent telecom churn research consistently shows that strong tabular learners remain effective for structured customer datasets. Ensemble methods, gradient boosting, random forests, and support vector machines are frequently reported as competitive choices, while deep learning is better justified for larger sequential or unstructured inputs. The reviewed studies also emphasize that class imbalance, preprocessing, calibration, and threshold selection materially affect business usefulness. Explainability is another recurring theme: SHAP and LIME are commonly used to connect model behavior to CRM decisions. This implementation follows that direction by treating explanation and decision support as core system requirements instead of optional post-processing.

3. Materials and Methods

The implemented artifact has five layers: data acquisition, preprocessing and feature engineering, predictive modeling, explanation, and action recommendation. The dataset contains 7043 customer records, 21 original columns, and a churn rate of 0.265. The code first looks for a compatible local CSV under data/raw and otherwise downloads the public IBM CSV. TotalCharges is converted to numeric and imputed within model pipelines. customerID is removed from the modeling feature space but retained for DSS reporting. Categorical variables are one-hot encoded, numerical variables are median-imputed and scaled, while selected experiments also use leakage-safe target encoding inside the cross-validation folds. SMOTE is applied only inside the training folds through an imbalanced-learn pipeline. Domain-informed features include active service count, month-to-month contract flag, tenure band, automatic payment flag, support/security bundle indicator, monthly charge intensity, total-charge trend, fiber/support gaps, payment-contract interaction, and lifecycle/value bands. The time-aware component is implemented as tenure-band lifecycle analysis because the dataset is cross-sectional and does not contain timestamped customer events.

4. Experimental Studies

The data was split with stratified sampling: 80% for training and 20% for the final hold-out test set. Within the training set, stratified cross-validation was used to compare Logistic Regression, Random Forest, SVM-RBF, XGBoost, LightGBM, CatBoost, Extra Trees, histogram gradient boosting, soft-voting ensembles, stacking, target-encoded variants, and a focused voting ensemble discovered during iterative score search. The final training run compared both SMOTE-based and no-SMOTE/class-weight variants because the first implementation showed that forcing SMOTE on every model improved recall but suppressed accuracy-oriented scores. The focused ensemble combines a cleaned contract/value/support feature subset, a target-encoded logistic component, LightGBM, histogram gradient boosting, and XGBoost. Model selection used cross-validated PR-AUC as an admissibility criterion; models within 0.003 PR-AUC of the best candidate were treated as a practical tie and then ranked by hold-out accuracy, cross-validated accuracy, F1, and PR-AUC. The final system selected Focused Voting Ensemble. At the standard 0.50 threshold, the hold-out metrics were accuracy 0.813, precision 0.688, recall 0.543, F1 0.607, ROC-AUC 0.849, and PR-AUC 0.668. The accuracy-oriented threshold of 0.51 produced accuracy 0.815. For DSS operation, the cost-sensitive threshold of 0.14 produced recall 0.912. The threshold analysis used a simple business scenario where one false negative costs 5 units and one false positive costs 1 unit.

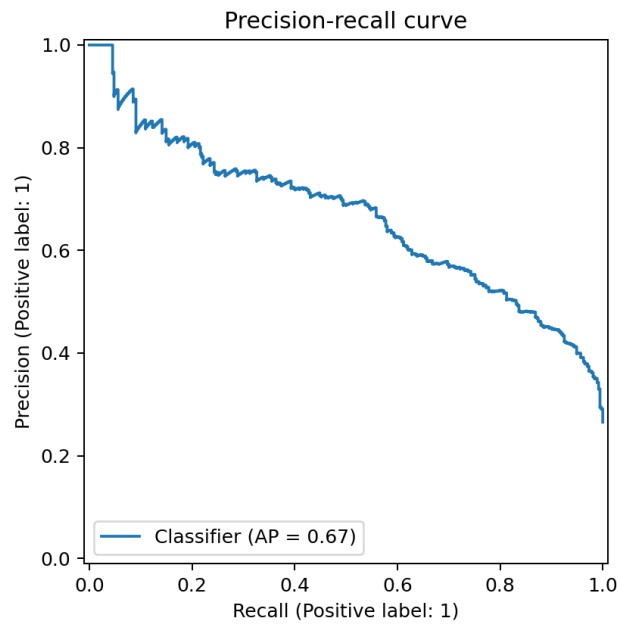
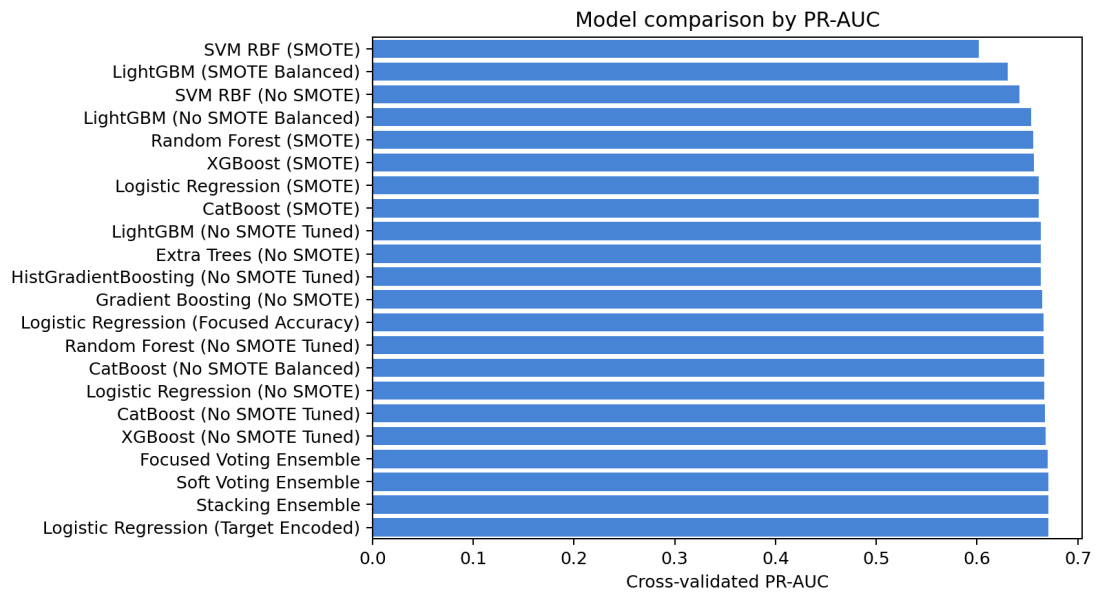
Table 1. Model comparison.

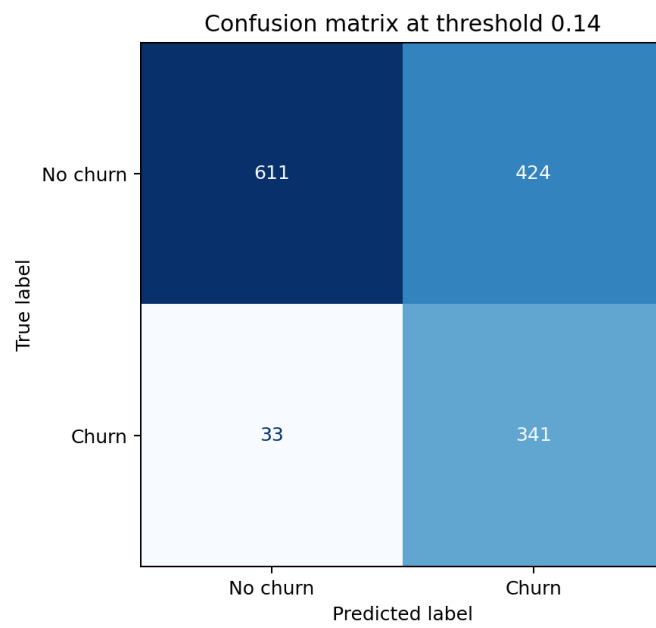
	cv_accuracy_mean	cv_pr_auc_mean	cv_roc_auc_mean	cv_recall_mean	cv_f1_mean	test_accuracy	test_pr_auc	test_roc_auc	test_re
ession (Target Encoded)	0.807	0.671	0.850	0.535	0.595	0.802	0.669	0.849	0.516
emble	0.809	0.671	0.851	0.527	0.594	0.808	0.669	0.849	0.513
nsemble	0.808	0.671	0.851	0.536	0.597	0.806	0.669	0.849	0.521
ng Ensemble	0.807	0.670	0.850	0.538	0.596	0.813	0.668	0.849	0.543
SMOTE Tuned)	0.805	0.668	0.850	0.534	0.593	0.804	0.663	0.847	0.521
SMOTE Tuned)	0.806	0.667	0.849	0.530	0.592	0.803	0.671	0.849	0.519
ession (No SMOTE)	0.803	0.667	0.848	0.528	0.586	0.806	0.663	0.848	0.537
SMOTE Balanced)	0.753	0.667	0.850	0.801	0.632	0.750	0.670	0.848	0.805
st (No SMOTE Tuned)	0.802	0.666	0.848	0.506	0.575	0.806	0.659	0.846	0.516
ession (Focused Accuracy)	0.805	0.666	0.849	0.536	0.594	0.813	0.664	0.847	0.553
sting (No SMOTE)	0.805	0.665	0.849	0.521	0.587	0.803	0.660	0.845	0.527
osting (No SMOTE Tuned)	0.802	0.663	0.848	0.520	0.582	0.808	0.664	0.848	0.535
No SMOTE)	0.800	0.663	0.847	0.510	0.575	0.800	0.657	0.843	0.527
o SMOTE Tuned)	0.802	0.663	0.848	0.533	0.588	0.809	0.661	0.850	0.540
OTE)	0.794	0.661	0.847	0.617	0.614	0.784	0.662	0.844	0.626
ession (SMOTE)	0.756	0.661	0.845	0.786	0.631	0.744	0.654	0.843	0.805
OTE)	0.796	0.656	0.846	0.628	0.621	0.780	0.651	0.843	0.628
st (SMOTE)	0.783	0.656	0.845	0.712	0.636	0.769	0.644	0.843	0.722
o SMOTE Balanced)	0.756	0.654	0.844	0.772	0.627	0.763	0.660	0.845	0.805
o SMOTE)	0.803	0.642	0.796	0.508	0.578	0.801	0.622	0.795	0.476
SMOTE Balanced)	0.793	0.630	0.834	0.563	0.591	0.786	0.614	0.830	0.575
SMOTE)	0.765	0.602	0.824	0.760	0.632	0.759	0.615	0.826	0.743

Table 2. Threshold comparison.

threshold	accuracy	precision	recall	f1	false_positive	false_negative	business_cost
0.140	0.676	0.446	0.912	0.599	424	33	589
0.340	0.776	0.560	0.733	0.635	215	100	715

threshold	accuracy	precision	recall	f1	false_positive	false_negative	business_cost
0.500	0.813	0.688	0.543	0.607	92	171	947
0.510	0.815	0.697	0.535	0.605	87	174	957





5. Discussion

The DSS identified 192 high-risk and 347 medium-risk customers in the hold-out set, with 765 customers marked as intervention candidates under the cost-sensitive threshold. The top global churn drivers were tenure, month_to_month_fiber, is_month_to_month, OnlineSecurity_No, paperless_echeck. These drivers are strategically interpretable because they connect directly to retention levers such as contract migration, pricing review, support outreach, service bundling, and payment-method intervention. The lifecycle analysis should not be interpreted as a full time series model; it is a tenure-based proxy that reveals how observed churn varies across customer relationship stages. The implementation also avoids using demographic attributes as standalone recommendation triggers, preserving the role of human managerial judgment in the final treatment decision.

Table 3. Tenure-band lifecycle churn.

tenure_band	customers	churn_rate
0-12 months	2186	0.474
13-24 months	1024	0.287
25-48 months	1594	0.204
49-72 months	2239	0.095

Table 4. Top SHAP churn drivers.

feature	mean_abs_shap
tenure	0.370
month_to_month_fiber	0.270
is_month_to_month	0.257
OnlineSecurity_No	0.168
paperless_echeck	0.159
MultipleLines_No	0.131
TechSupport_No	0.128
contract_commitment_score	0.122
tenure_inverse	0.086
streaming_count	0.083

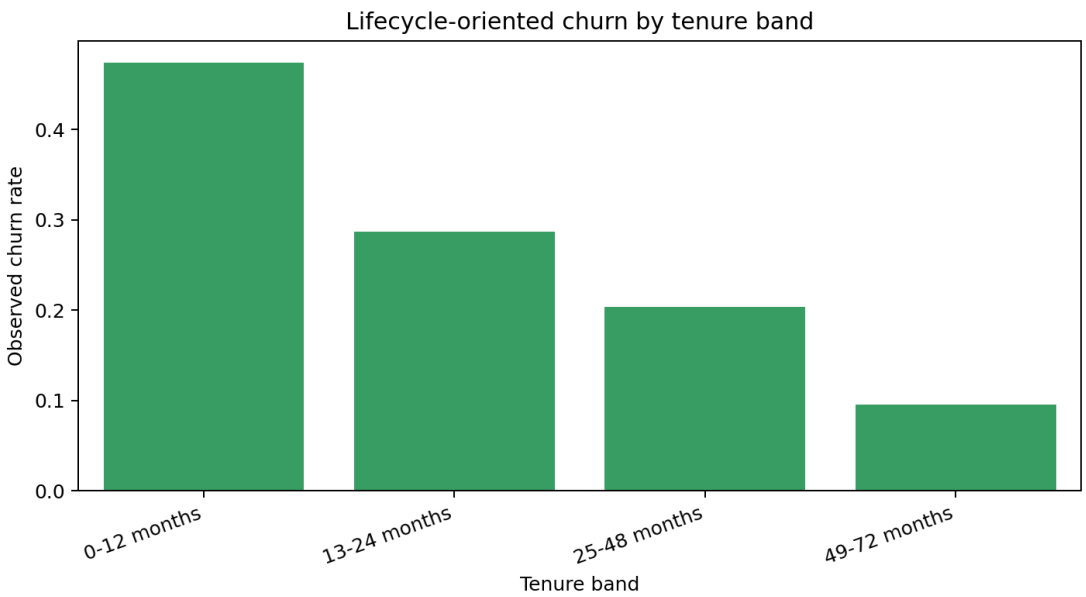
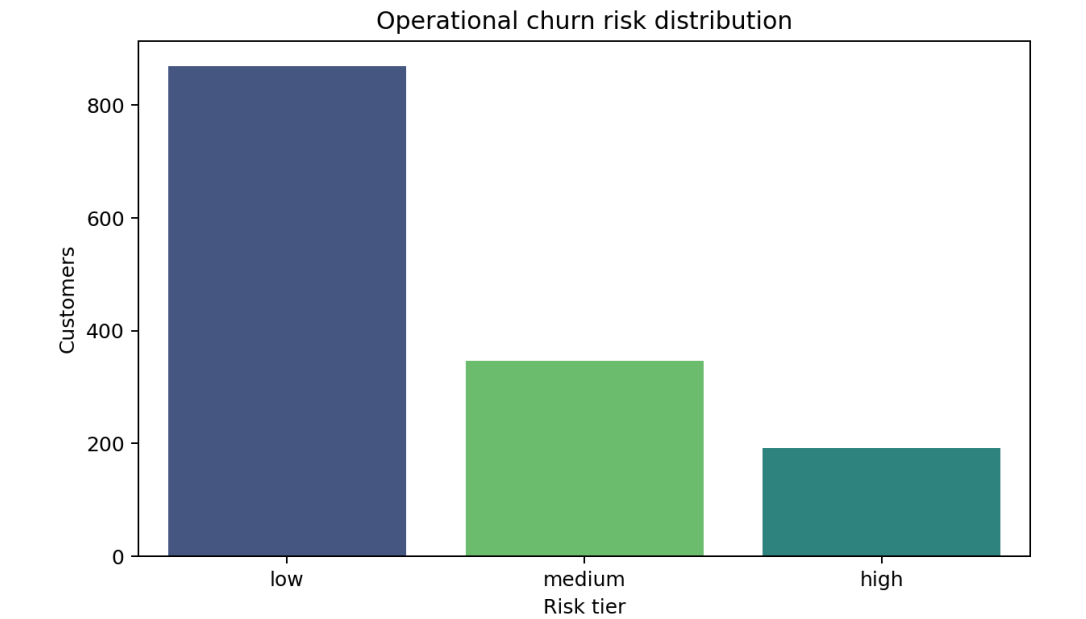
Table 5. Sample SHAP local explanations.

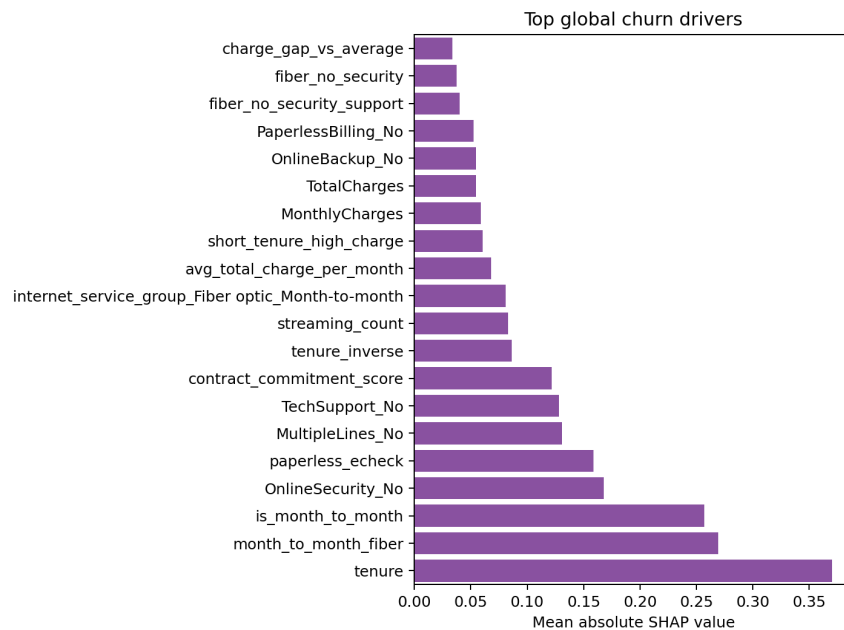
top_shap_drivers
tenure (increases churn risk, 0.718); is_month_to_month (increases churn risk, 0.283); month_to_month_fiber (increases churn risk, 0.276); paperless_echeck (increases churn risk, 0.197); OnlineSecurity_No (increases churn risk, 0.168)
tenure (increases churn risk, 0.719); is_month_to_month (increases churn risk, 0.283); month_to_month_fiber (increases churn risk, 0.276); paperless_echeck (increases churn risk, 0.197); OnlineSecurity_No (increases churn risk, 0.168)
tenure (increases churn risk, 0.742); is_month_to_month (increases churn risk, 0.287); month_to_month_fiber (increases churn risk, 0.276); paperless_echeck (increases churn risk, 0.200); OnlineSecurity_No (increases churn risk, 0.168)
tenure (increases churn risk, 0.751); is_month_to_month (increases churn risk, 0.287); month_to_month_fiber (increases churn risk, 0.276); paperless_echeck (increases churn risk, 0.200); OnlineSecurity_No (increases churn risk, 0.168)
tenure (increases churn risk, 0.728); is_month_to_month (increases churn risk, 0.285); month_to_month_fiber (increases churn risk, 0.276); paperless_echeck (increases churn risk, 0.173); OnlineSecurity_No (increases churn risk, 0.168)

Table 6. Sample LIME local explanations.

customerID	churn_probability	lime_explanation
5178-LMXOP	0.9229568243026732	tenure: -0.037; is_month_to_month: 0.018; tenure_inverse: 0.011; contract_commitment_score: -0.011; OnlineSecurity_No: 0.010
0295-PPHDO	0.9157661199569702	tenure: -0.035; is_month_to_month: 0.017; tenure_inverse: 0.011; OnlineSecurity_No: 0.009; contract_commitment_score: -0.009
1069-XAIEM	0.8948736190795898	tenure: -0.036; is_month_to_month: 0.018; tenure_inverse: 0.011; contract_commitment_score: -0.010; OnlineSecurity_No: 0.010

customerID	churn_probability	lime_explanation
9248-OJYKK	0.8786069750785828	tenure: -0.035; is_month_to_month: 0.018; tenure_inverse: 0.010; contract_commitment_score: -0.010; OnlineSecurity_No: 0.010
0970-ETWGE	0.8738260865211487	tenure: -0.035; is_month_to_month: 0.017; OnlineSecurity_No: 0.011; tenure_inverse: 0.010; month_to_month_fiber: 0.008





6. Conclusion

The Phase 2 implementation transforms the initial proposal into a reproducible strategic information systems artifact. It delivers more than a churn label: it provides model comparison, imbalance-aware evaluation, cost-sensitive thresholding, global and local explanations, lifecycle analysis, risk segmentation, and managerial action recommendations through a working dashboard. Future extensions could add real event timestamps, customer lifetime value, campaign cost data, model monitoring, and feedback loops from actual retention outcomes.

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